Pandemic Effect on Education System among University Students

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Abstract. Online exam systems are pieces of software used in online education to evaluate students' performance. However, both the security of the system they respond to and a broad understanding of the risks they may face must be addressed. In this article, we will look into the most recent issue, which is online exams and their security systems. Because most students depended on Google, their friends, and other resources for help, we focused on software-based security and online sup-port for cheating like IP address tracing, block local file accessing etc. We did this project step by step like collecting data, data pre-processing, data cleaning, training data and trying to make a prediction of how the student cheated the most. Then we can start our future work as we want to increase the security of online exams. According to the findings of a few study papers we read, individuals that use e testing do so for all main kinds of assessment, including quizzes, midterms, and final exams. The most common format for these tests is multiple-choice, which is certainly the easiest to cheat on, but the faculty admits they aren't routinely employing anti-cheating tactics. So, this research paper will help the new world in the field of online exam systems.

Keywords: E-learning, Online exam, Machine learning, KNN, Naïve Bayes, Logistic regression.

1 Introduction

Education has not been spared as technology has rapidly digitized all services and offerings, bringing them to the web platform. The pervasiveness of technology has facilitated a seamless shift to the online environment, thanks to ubiquitous high-speed internet and computers. Learning Management Systems, which are software applications that track, report, administer, and document materials shared with students, have advanced and adapted in universities, schools, and other educational institutions. This teaching and evaluation technique allows for a balanced approach to knowledge transmission as well as easy candidate grading. The goal of online exams is for assessment givers to be able to make the transition from offline to online processes. The online market is constantly expanding, and factors including convenience, scalability, expanded reach, and customization are influencing its adoption. Established evaluation methodologies are rapidly approaching saturation and will become obsolete in the near future. Exams are designed to fairly assess a student's ability. The exam results would be erroneous in representing their true competence if the exam's impartiality was compromised. Cheating raises various difficulties that we as an educational institution must be aware of, in addition to the incorrect assessment of students' learning progress. It's no longer necessary to scribble math formulas on your palms. Nowadays, students have devised ingenious techniques to cheat, most of which take advantage of the technology available to them. Sharing student's screen with another computer, utilizing cutting edge technological devices, using mobile apps and taking notes on a smartphone, creating false identities in order to obtain third-party assistance etc. Exam proctoring is the most prevalent solution. It is specifically developed to allow students to take online tests safely, with features like auto-mated monitoring, ID verification, and machine lockdown, as well as IP address tracing and blocking local file access. This option, however, may be fraught with technical difficulties and significant expenditures. But, in order to discover a solution, we must first determine which elements are appropriate for pupils to cheat on. In this study work, my team and I used machine learning to forecast how students cheated the most and how they could improve their grades. We use three different sorts of algorithms in our project. To accurately train our data, we used Logistic Regression, KNN, and Naive Bayes. We test 20% of the data first, then train the remaining data our project's core theme is fairly straightforward. The title encapsulates the entire viewpoint. Though it is a recent issue, we need to enhance the security mechanism of online tests. Students employ a variety of strategies to pass. Students, on the other hand, constitute a country's backbone. As a result, we must give it considerable consideration. As a result, our initiative first identifies a problem before attempting to address it. As previously stated, it is a new trend, so just a few people are beginning to work with it. As a result, we were unable to locate any datasets suitable for our investigation. We manually collect data using a Google form. We put certain qualities there, and individuals entered data using these attribute questions. We gather approximately 1,000 data points, transform them to numerical values, train them, and then write our research using the trained data. Better and safer online exams would be provided by a secure online examination system. The whole process and quality of the exam process can be hampered by online examination systems that lack sufficient information security protocols. Our future work has a high level of ambition. We create a low-cost, practical solution that will benefit all types of students and educational institutions.

2 Review Works

Giustolisi, et al. (2013) aims to provide a research roadmap in the field. Safe, verifiable, and accountable e-exam systems have identified three separate study goals, each of which is led by particular research issues. They've just suggested a safe online examination method there [1].

Shraim, et al. (2019) despites the fact that higher edu-cation institutions in Palestine are increasingly using online exams, the perspectives of students on their use remain unexplored. As a result, the major focus of this study is to look into the online testing processes used by higher education institutions from the perspective of students who have taken online 2 exams at Palestine Technical University-Kadoorie (PTUK). Its findings will assist the institution in identifying essential and strategic components of effective E-exam design and administration to support students in higher education learning [2].

Muzaffar, et al. (2021) develops a review procedure that entails six phases according to the standard SLR standards. The first two phases are already addressed in the introduction to the article, and the next four phases are detailed in the accompanying segments [3].

Sarrayrih, et al. (2016) creates a completely automated Online Exam System that can gather, combine, and analyze data while also evaluating the program's impact. The system will only be accessible to authorized users with certain duties (Administrator: maintains the website, Institutes: register to administer examinations, Students: give exams online) [4].

Ramu, et al. (2013) states because of the high stakes involved in online exams, reliable student authentication is highly important. Authentication verifies that the users are who they claim to be. Authentication is used to verify the identity of online students and is a critical component of security in an online test setting [5].

Alotaibi et al (2010) to protect templates against fraudulent use, he recommended utilizing a distorted signal or feature vector. Using data hiding and watermarking techniques, fingerprint images can be made more secure by detecting changes and hiding one biometric within an-other. To protect against intruders, the scanned fingerprints will be stored in encrypted form. However, the time between fingerprint scans may be so short that it creates storage issues for such a large amount of data. As a result, the time interval can be increased to at least three seconds, allowing for a reasonable reduction in data volume [6].

Feinman et al (2018) according to this research article, a quasi-experimental study was conducted with the primary objective of determining the effectiveness of non-biometric security measures. Using a taxonomy of cheating reduction approaches based on the fraud triangle hypothesis, the security measures were selected for use. According to this study, the fact that the number of students who replied to all test items was higher on the proctored tests than on the proctored exams may imply that students tend to skip more questions in proctored contexts than they do in proctored environments Despite this, the sample was varied in terms of majors and included typical community college students. In addition, the sample of students in the study was not drawn at random from the population of all community college students who take introductory statistics in web-assisted environments. Whatever the case, the application of the security methods employed in the study may make it possible to assess student knowledge in a credible, low-cost, and convenient manner [7].

Apampa et al (2010) investigation into the current user security paradigm and the vulnerabilities that it contains is the subject of this study. According to the authors, a paradigm for user security in summative electronic evaluations has been developed specifically for this situation. Data security, location security, software security, and the network security of summative e-assessments are some of the issues to be concerned about. Instead of categorizing all impersonation threats as Type A, Type B, or Type C, the authors of this research categorize them into three types: A, B, and C. The tracking module computes the difference between two video frames in order to determine the location and position of the learner on the track. Student risk levels are determined at various points during a test using information from the tracker, which is fed into a classifier module. The current model's identification and authentication security goals are insufficient in terms of protecting student information from impersonation attacks [8].

Lin et al. (2004) above study analyses how tool creation can work in conjunction with security to provide a learning environment that is safe, efficient, and effective. They detailed various different e-learning apps and categorized them according to their functions, which they then listed. They stated the security and privacy needs in terms of the situations and the types of data shared, based on this categorization. In addition, they planned to upgrade our e-learning systems in order to make them comply with the new distance education standard, known as the Sharable Content Object Reference Model (SCORM). They worked on student performance evaluation, Course Authoring and Student Assessment Application, Distributed Multimedia Presentation Application with Floor Control, Security and Privacy: Requirements and Solutions, Copyright Protection, and Floor Control Security while at this location [9].

Weiner et al. (2017) aim of the research seems to be to compare online remote proctoring of high-stakes licensure examinations to traditional test center proctoring of the same examinations in order to determine whether or not these alternative test administration conditions are equivalent or whether or not they have potential differences. The findings of the study suggested that examinee scores achieved under online kiosk-based proctoring conditions were psychometrically sound and equivalent to the identical exams taken under traditional test center proctoring circumstances. Furthermore, examinees assessed online kiosk-proctored examinations well, and ratings of testing conditions were essentially uncorrelated with exam performance, implying that examinees' experiences under the various testing conditions had no impact on their performance on the exams. Globally speaking, the study's findings endorsed the use of a kiosk-based remote proctoring approach for high-stakes test delivery that was equivalent to using an onsite test center proctoring method. Despite this, there was no indication that remote proctoring had any effect on test results [10].

Saleh et al. (2014), in this study, author present a variety of security concerns in computer-based and online testing. In addition, they investigated a variety of authentication techniques for examinee authentication. Furthermore, they suggested a novel authentication system that combines the username/password with palm-based biometrics. The proposed scheme will enhance the authentication process up to an optimum level during computer-based or online-based testing. The video capturing technique should also be merged with the existing system, in order to make it more secure in the future. Also, the CATSIM simulation tool should be used to simulate the entire authentication and verification process. The suggested system should improve the authentication procedure during computer-based or online testing to an optimal level. The video capture technology should also be integrated into the existing system in the future to make it more secure. In addition, the CATSIM simulation tool should be utilized to model the whole authentication and verification process [11].

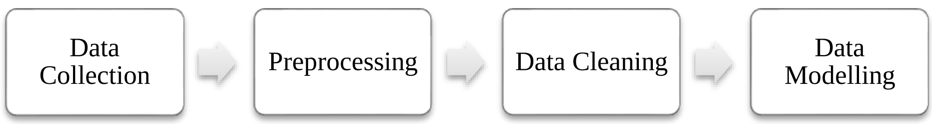
Im Y. Jung et al. (2009), this author shows how the SeCOnE system provides both secure online exam management and an e-monitoring method for the prevention and detection of cheating. The strategies described in this article for preventing and detecting cheating encompass cheating methods known for the online test procedure through computer or Internet, although they may not address all conceivable cheating methods. This paper is intended for tests conducted over the Internet at a defined time with a single problem set, but with no restrictions on testing location. The fact that the second can be applied to exams given at various periods is a strong feature. To prevent cheating during the test, the examiner should prepare as many problems sets as there are exam hours. One overhead expense for this system is the preparation of the equipment should be used to monitor and authenticate the entities, such as Webcams and microphones. Another overhead to consider is the net burden caused by monitoring data transport and storage, but this is not a big issue when data compression is utilized and more monitor servers are provided. They should look at the consequences of hostile entities in the system, as well as the procedures involved in dealing with failures to improve this paper. The author should do some more work for improving this paper for enhancing security for online exams using group cryptography [12].

Mahantesh Mathapatiat et al. (2017), this proposed technique increases the level of security for conducting online tests. The primary concept of this approach is the incorporation of live video and real tokens held by the user. This study demonstrates the viability of the results by testing a fully functional model. Through an evaluation of time, porn rate, and effort, it is demonstrated that the novel technique performance corresponds to the achievement level in traditional. The new approach is strong since it can survive any unfavorable circumstance of severe testing. The new method's performance is compatible with the standards and delivers steady, consistent results, making the new graphical password scheme reliable. When compared to previous approaches, the data show that the new methodology considerably enhances confrontation to attacks. Thus, they should improve their paper as well as system to meet the criteria of online examination systems. Finally, the new system secures and ensures the highest level of security during examinations. It can be improved by using compound character and image size [13].

Alex et al. (2019) the aims and research question were justified and discussed throughout the literature review, and the findings were afterwards compared and contrasted to the findings from the semi-structured interviews. The research topic is: How can institutions assure proper student identity when examinations are taken outside of school by using various authentication methods? was examined using a variety of authentication techniques Institutions must keep up with what is going on across the world, which is justified by the rapid advancement of technology and potential applications of machine learning and artificial intelligence in e-learning. As observed throughout the literature study, potential outcomes and online tests have previously been introduced; however, security safeguards and fully supervised services may be lacking. Students need test time to focus on what they're doing, not to switch off and deal with any issues with third-party monitoring providers, for example. Unethical behavior is a problem that has been handled at institutions where tests are held on-site and hence does not pose a greater risk while working online. They conducted an empirical study on a group of online students from local and overseas Universities. The result shows the impact of questions type on the usability, in particular the amount of time taken by the introduction of the proposed approach. They should conduct a post experiment survey to collect student feedback on the proposed technique [14].

Jerry Chun Wei Lin et al. (2016) a plethora of methods has been developed in this paper to anonymize relational data. However, preventing the leakage of sensitive information in transactional data is also a key concern. Algorithms suggested for anonymizing transactional data often result in significant information loss. In this work, they introduce PTA, a unique anonymization system composed of three modules. It anonymized transactional data with little information loss and is also highly fast when compared to state-of-the-art transactional data anonymization techniques. Significant tests should be conducted to evaluate the performance of the developed system against state-of-the-art techniques for anonymizing transactional data in terms of runtime and information loss. The outcome of this work is quite impressive and also it outperforms the existing handcrafted feature extraction methods [15].

3 Research Methodology



3.1 Data Collection

The authors are making use of the real-time data set that was acquired through a Google form on their website. The form was distributed on social media, and from there, all of the necessary information was gathered. The dataset contains information about students' online exam experiences, including whether or not they received any online assistance. There were some questions about the dataset. Questions are:

- You're Grade in Online Examination System.
- Get help in exams from your classmates.
- Get help in exam from Google and YouTube
- Friend circle grouping exam
- Get help from any other source?

Answer to the first question was two; the first one was increased, and the second one was de-creased, resulting in a total of two answers. The responses to the second question were given in percentages. The answer to the third question is the same as the answer to the second question. The fourth and fifth questions have two possible answers: yes and no. The final question had four responses, which were: electricity problem, network problems, Device Management problem, and others.

3.2 Preprocessing

Following the collection of the data, the dataset was discovered to be in Microsoft Excel format. For the purposes of this project, data was first converted into a CSV format. Many nominal values were discovered after the conversion process was complete. Those nominal values are manually processed and converted to numerical values by a computer program. These values were then translated into integer numbers using a string conversion algorithm. Take, for example, the following:

- In Result
  - Increase: 1
  - Decrease: 0
- Online Help and Friend Help
  - Percentage deviation from typical numerical values.
- Group Exam and other source
  - Yes: 1,
  - No: 0
- Problem During exam
  - Electricity Problem: 1
  - Net Problem: 2
  - Device Management Problem: 3
  - Others: 4

3.3 Data Cleaning

Following the cleaning of the data, we proceed to the identification of the dependent and independent variables. In this particular instance, we have chosen X and Y. Y is the dependent variable, whereas X is the independent variable in this situation. Following that, we selected 20 percent of the data to train on and 80 percent of the data to give accuracy for the model.

3.4 Data Modelling

3.4.1 K Nearest Neighbor

KNN is an abbreviation for K-Nearest Neighbors, which is a type of proximity search. A learning algorithm is used, and it is under the supervision of a trained individual. It establishes the distance between each row of training data and each row of test data using the distance function. By their respective values, arrange the distances determined in ascending order. The first k rows of a sorted array are returned by this function. In this row, find out which class is the most frequently seen. Should be returned as the projected class.

$$Pr\ (Y\ =\ j|X\ =\ x0)\ =\ 1k \sum i\ \in\ N0(yi\ =\ j)\ \dots\ \dots\ \dots\ (i)$$

3.4.2 Logistic Regression

Machine learning algorithm logistic regression can be used to predict if an observation belongs to one of two classes based on its characteristics. In financial analysis, linear regression is a machine learning technique based on super-vised learning. It's in charge of carrying out a regression operation. These models use independent variables to predict the value of a prediction. I linear regression is used to predict the value of a dependent (y) variable given the value of an independent I (x).

$$Log\ (\ 1\ -\ p\ /\ )\ \dots\ \dots\ \dots (ii)$$

3.4.3 Gaussian Naïve Bayes

A simple learning algorithm, Naive Bayes makes use of the Bayes rule in con-junction with a strong assumption that the attributes are conditionally independent, given the class in which they are used. It is possible to build a Bayesian classifier using a probabilistic model in which the classification is represented by a latent variable that is probabilistically related to the observed variables.

$$P(A|B) = P(B|A) \cdot P(A)\ P(B) \dots\ \dots\ \dots\ (iii)$$

4 Result & Discussion

4.1 Confusion Matrix

The values of the KNN, Logistic, and Naive Bayes Confusion Matrix are listed in this table. Here the formula of this Confusion Matrix is True positive 3 (TP), True negative (TN), False positive (FP), False negative (FN).

Table 1. Confusion Matrix

Algorithms	**Predict**		
KNN	174	14	Actual
	6	4	
Logistic Regression	178	0	
	22	0	
Gaussian Naïve Bayes	178	0	
	22	0	

4.2 Classification Report

A Classification report must be created to evaluate the accuracy of the predictions provided by a classification algorithm. Classification metrics precision, recall, and f1-score are displayed in a single report row for each class. Metrics like true positives and false negatives are factored into the formula for calculating performance. A comparison table has been created with both the decreased and raised values. The algorithm's precision is also taken into account.

Table 2. Classification Report

Algorithm	Class	Precision	Recall	F1-Score	Accuracy (%)
KNN	Decreased	0.80	0.73	0.76	
	Increased	0.97	0.98	0.97	95
Logistic Regression	Decreased	0	0	0	
	Increased	0.89	1	0.94	89
Naïve Bayes	Decreased	0	0	0	89
	Increased	0.89	1	0.94	

4.3 Accuracy

The machine learning model measures the accuracy with which a model finds correlations and patterns between variables in a dataset based on the input data, or training data. This accuracy chart illustrates the relative importance of KNN, Logistic, and Naïve Bayes methods. And KNN had the highest Accuracy rating, earning a score of 95%.

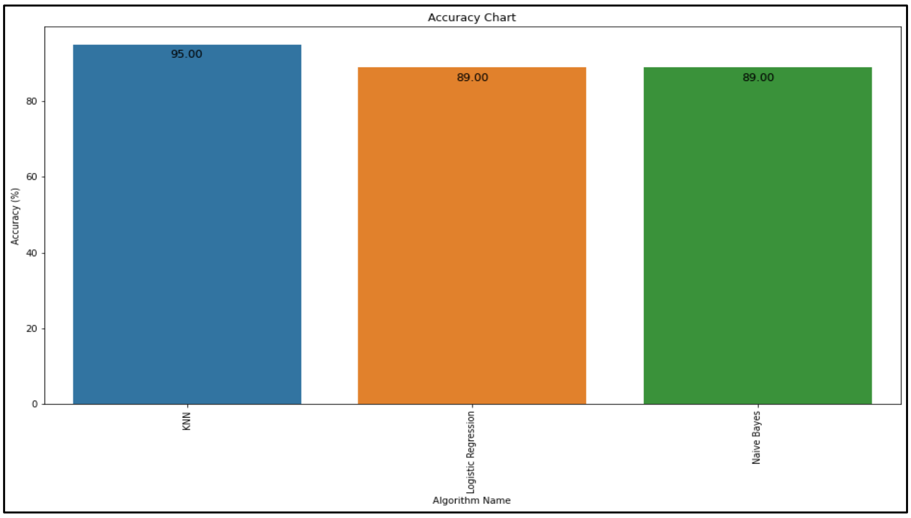


Fig. 1: Accuracy Chart

5 Comparative Analysis

An online examination system makes it possible to prepare and deliver exams entire-ly online. Because of internet-based configuration options, this is a real possibility. This paper's goal is to make the online test system more secure. K Nearest Neighbors, Logistic Regression, and Gaussian Nave Bayes were a few of the confusion matrices we used in our research. The classification report is summarized in the table 2.
On a 95% of all occasions, the KNN matrix performed flawlessly.
There were two methods used to achieve the same level of accuracy: logistic regression and Gaussian Naive Bayes. The precision of the results used by different authors and researchers, on the other hand, has led to conflicting conclusions. Online exams can be made more secure by utilizing biometric authentication methods such as fingerprint recognition. Through the use of online testing methods, the objectivity of evaluations can be improved while the effort required to make corrections is reduced. For new student orientation and registration, this is very useful when utilized in large classrooms (STEOP). Modern assessment appears to be essential for modern education at this time. The data for this report was gathered using a google form. Often times, authors may gather their information from a variety of various websites all over the internet. This leads me to believe that updating exam security using real data is preferable in this instance.
Students who use e-testing for quizzes, midterms, and exams of all kinds, according to research after research.
However, instructors report they do not regularly use procedures to avoid academic dishonesty despite the most common exam format being multiple choice, which makes cheating the easiest. The WebCT courseware platform offers numerous preventative measures for educators that are as easy to implement as using good judgement and selecting a single directive. If academic dishonesty is an issue, more should be done to prevent it for obvious reasons. There isn't a solution in adding more work to the faculty; rather, swift decisions about basic options and responsibilities must be made

6 Acknowledgment

First and foremost, we would like to express our profound gratitude to Allah for His tremendous grace, which has enabled us to complete this work successfully. We also express our gratitude to my respectable course instructor. Because he is with us throughout our whole journey. Finally, we'd want to thank everyone who helped us, including well-wishers, friends, family, and elders. In addition to concepts. This study is the result of a great deal of hard work.

7 Future Scope & Conclusion

This paper explains how our online test system will be safer and more secure, as well, as how students might improve their grades by cheating online. The measures presented in this work for preventing and detecting cheating cover cheating methods identified for online exam processes through computer or Internet, however, they may not cover all possible 95% 89% 89% Accuracy Chart KNN Logistic Regression Naïve Bayes 4 cheating methods. Our primary goal is to improve the examination atmosphere. We will take things one-step at a time. To begin, this study will investi-gate how students cheated, as well as the difficulties they encountered during the online exam. Following that, we will address their concerns as well as the subject of cheating. As a result, the online testing method improved day by day. We concentrated on software based security and their internet assistance for cheating because most students relied on Google, their friends, and other resources for aid. That is why we concentrate mostly on that aspect. To classify our paper's intention, we employ KNN, Logistic Regression, and Naive Bayes algorithms in our coding section. In computer based and online-based testing, we have looked into many security issues. We have also looked into a number of other authentication mechanisms for examinees. Finally, the lack of in-person supervision owing to geographical limits, along with accessible technologies, has made it easier for students to cheat on online tests. Knowing how students cheat in novel ways can help you find the proper answer and alleviate security concerns.

In the future, we intend to improve our projection by incorporating more and deeper records from our own database, which we are continuously building with increasingly complex data. We would like to incorporate more security solutions, as well as how to create them from a Bangladeshi standpoint. We currently do not have enough dataset; but, in the future, we will have a large dataset with many variables that will indicate how we should improve security in the online examination system. The con-sequences of hostile entities in the system, as well as the mechanisms involved in handling errors, will be the focus of future research.

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